

PureBasic OOP Reference Manual (English)

Welcome to the official documentation for the PureBasic Object-Oriented Programming transpiler and framework.

1. Introduction

PureBasic OOP brings high-level Object-Oriented syntax while retaining the native speed, lightness, and power of the PureBasic compiler.

Key Features:

- **Classes & Methods**: Clean declarations using ``Class`` and ``Method``.
 - **Encapsulation**: Access control for fields and methods (``Public``, ``Protected``, ``Private``).
 - **Constructors & Destructors**: Automatic instantiation (``New_*`) and explicit memory deallocation (``Free()``).
 - **VTable Polymorphism**: Transparent mapping to PureBasic ``Interface`` and virtual function tables (``DataSection``).
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2. Syntax & Object Grammar

Class Declaration

```
```oop
Class Chien
 Private nom.s
 Private age.i

 Public Method Init(nom_p.s, age_p.i)
 Public Method Aboyer()
 Public Method Free()
EndClass
```
```

Method Implementation

Each method accesses instance data using the ``This`` reference pointer.

```
```oop
Method Chien::Init(nom_p.s, age_p.i)
 This\nom = nom_p
 This\age = age_p
EndMethod

Method Chien::Aboyer()
 PrintN(This\nom + " dit : Wouaf ! Wouaf ! (Age : " + Str(This\age) + " ans)")
EndMethod
```
```

3. Generated PureBasic Code Plumbing

The transpiler converts OOP constructs into native, compilable PureBasic code:

```
```purebasic
Interface Chien_vt
 Aboyer()
 Free()
EndInterface

Structure Chien_Inst
```

```
 *VTable.Chien_vt
 nom.s
 age.i
EndStructure

Procedure Chien_Aboyer(*This.Chien_Inst)
 OpenConsole()
 PrintN(*This\nom + " dit : Wouaf ! Wouaf ! (Age : " + Str(*This\age) + " ans)")
EndProcedure

Procedure Chien_Free(*This.Chien_Inst)
 FreeStructure(*This)
EndProcedure

DataSection
 Chien_VTable_Data:
 Data.i @Chien_Aboyer()
 Data.i @Chien_Free()
EndDataSection

Procedure.i New_Chien(nom.s, age.i)
 Protected *obj.Chien_Inst = AllocateStructure(Chien_Inst)
 If *obj
 *obj\VTable = ?Chien_VTable_Data
 *obj\nom = nom
 *obj\age = age
 EndIf
 ProcedureReturn *obj
EndProcedure
```
```

4. Compilation Guide

To compile a generated PureBasic source file into a console executable:

```
```cmd
"C:\Program Files\PureBasic\Compilers\pbcompiler.exe" "src/test_chien_generated.pb" /CONSOLE /EXE "src/test_chien.exe"
```
```